

Year	Return
9	5.5%
10	4.0%

5. The fund's arithmetic mean return over the 10 years is *closest* to:

- A. 2.97 percent.
- B. 3.00 percent.
- C. 3.33 percent.

Solution:

B is correct. The sum of the returns is 30.0 percent, so the arithmetic mean is $30.0\%/10 = 3.0\%$.

6. The fund's geometric mean return over the 10 years is *closest* to:

- A. 2.94 percent.
- B. 2.97 percent.
- C. 3.00 percent.

Solution:

B is correct. The geometric mean return is calculated as follows:

$$\bar{R}_G = \sqrt[10]{(1 + 0.045) \times (1 + 0.06) \times \dots \times (1 + 0.055) \times (1 + 0.04)} - 1,$$

$$\bar{R}_G = \sqrt[10]{1.3402338} - 1 = 2.9717\%.$$

7. The harmonic mean return over the 10 years is *closest* to:

- A. 2.94 percent.
- B. 2.97 percent.
- C. 3.00 percent.

Solution:

A is correct. The harmonic mean is calculated as follows:

$$\bar{X}_H = \frac{n}{\sum_{i=1}^n (1/(1 + r_i))} - 1,$$

$$\bar{X}_H = \frac{10}{\left(\frac{1}{1.045}\right) + \left(\frac{1}{1.06}\right) + \dots + \left(\frac{1}{1.055}\right) + \left(\frac{1}{1.04}\right)} - 1,$$

$$\bar{X}_H = \left(\frac{10}{9.714}\right) - 1 = 2.9442\%.$$

Consider the box and whisker plot in Exhibit 10: